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// WATCH: ADB APP CONTROL: SIDELOADING APPS / UNINSTALLING APPS & FACES / SCREEN DPI

ADP APP CONTROL: https://adbappcontrol.com/en/

<Sideload Apps (E.G. Google Voice)>

To Sideload from Phone instead of PC https://www.youtube.com/watch?v=0k3v570ryLY

WATCH: Settings > Enable developers mode (Settings > About Watch > Software Info > "Software Version" 5Xs)

WATCH: Settings > Developer Options > Enable: Enable ADB Debugging / Debug Over WiFi / Wireless Debugging

PC: Connect Watch & PC to same WiFi

WATCH: Settings > Connections > WiFi Settings > Click the Network > Get IP Address: 192.168.77.77

PC: ADB App Control > Top Right Corner > Type in IP > Hit Connect button (to right of IP)... WATCH: when prompted "Always Allow Connection"

PC: Download Google Voice (or other) apk: https://google-voice.en.uptodown.com/android

PC: ADB App Control > Install (top right) > Drag APK & Drop onto Program Window > Rt-Click the App (e.g. Google Voice) > Install (will take a while)

(NOTE: Use phone's Phone # for setting up Google Voice on Watch)

WATCH: Settings > Developer Options > Disable: ADB Debugging

<Uninstalling Apps & Preinstalled Watch Faces>

[Automatically Uninstalling]

1. Enable developers mode: Watch Settings > About Watch > Software Info > "Software Version" 5Xs to turn on developer mode

2. Enabling ADB: Watch > Settings > Developer Options >
   Enable ADB Debugging
   Debug Over WiFI
   Enable Wireless Debugging

3. Connect PC & Watch to same WiFi... then: Watch > Connections > WiFi Settings > Click the Network > Get IP Address: xxx.xxx.xx.xx

4. WATCH: Make sure none of the stock watchfaces are loaded

5. PC: ADB App Control > Top Right Corner > Type in IP > Hit Connect button (to right of IP)
WATCH: when prompted "Always Allow Connection"

6. PC: ADB App Control > Search "watchface" > Select them (see notes below before disabling) > Disable (bottom right) > Apply

(NOTE: May be dangerous to uninstall instead of disabling - may not be able to roll back)
(NOTE: May be dangerous to uninstall those with Green Droid)

(NOTE: The following MAY NOT BE OKAY DONT UNISTALL):
Emergency Watch Face
Ticking Sound Setting
Watch Faces

(NOTE: The following are OK to uninstall, although they have a Green Droid instead of Watch Face Preview):
Basic Dashboard
Big Number
Info Brick
Live Wallpaper
Weather Center
All other watch faces that have a Watch Face Preview Icon

7. WATCH: Don't forget to turn off the 3 ADB Settings on watch when you're done (but disabling ADB will probably disable WiFi and Connection)

[Manually Uninstalling]

https://forum.xda-developers.com/t/debloat-galaxy-watch-4.4324147/

BATCH .bat SCRIPT: https://forum.xda-developers.com/t/debloat-galaxy-watch-4.4324147/post-87697011

You can Disable or Reinstall as well:
pm disable-user --user 0 (nameofpackage)
pm uninstall -k --user 0 (nameofpackage)
pm install-existing (nameofpackage)

Running ADB (https://forum.xda-developers.com/t/watch4-adb-commands-disable-enable-uninstall-restore-system-app-install-pull-apps.4324063/)
PC > Open adb minimal and write:
adb connect (IP)
adb shell done

Then all these

(WATCH FACES)
pm uninstall -k --user 0 com.samsung.android.watch.watchface.analogmodular
pm uninstall -k --user 0 com.samsung.android.watch.watchface.analoguefont
pm uninstall -k --user 0 com.samsung.android.watch.watchface.animal
pm uninstall -k --user 0 com.samsung.android.watch.watchface.aremoji
pm uninstall -k --user 0 com.samsung.android.watch.watchface.basicclock
pm uninstall -k --user 0 com.samsung.android.watch.watchface.basicdashboard
pm uninstall -k --user 0 com.samsung.android.watch.watchface.bespoke
pm uninstall -k --user 0 com.samsung.android.watch.watchface.bitmoji
pm uninstall -k --user 0 com.samsung.android.watch.watchface.digitalfont
pm uninstall -k --user 0 com.samsung.android.watch.watchface.digitalmodular
pm uninstall -k --user 0 com.samsung.android.watch.watchface.dualwatch
pm uninstall -k --user 0 com.samsung.android.watch.watchface.emergency
pm uninstall -k --user 0 com.samsung.android.watch.watchface.endangeredanimal
pm uninstall -k --user 0 com.samsung.android.watch.watchface.healthmodular
pm uninstall -k --user 0 com.samsung.android.watch.watchface.large
pm uninstall -k --user 0 com.samsung.android.watch.watchface.livewallpaper
pm uninstall -k --user 0 com.samsung.android.watch.watchface.myphoto
pm uninstall -k --user 0 com.samsung.android.watch.watchface.mystyle
pm uninstall -k --user 0 com.samsung.android.watch.watchface.simpleanalogue
pm uninstall -k --user 0 com.samsung.android.watch.watchface.simpleclassic
pm uninstall -k --user 0 com.samsung.android.watch.watchface.simplecomplication
pm uninstall -k --user 0 com.samsung.android.watch.watchface.superfiction
pm uninstall -k --user 0 com.samsung.android.watch.watchface.together
pm uninstall -k --user 0 com.samsung.android.watch.watchface.typography
pm uninstall -k --user 0 com.samsung.android.watch.watchface.weather

WANT TO BUT CAN'T DISABLE SAMSUNG PAY BECAUSE THEN SPayRemap_wear WONT WORK:

ADB APP CONTROL > Search "samsungpay" > Rt-Click > Disable

OR CONSOLE IN ADB APP CONTROL > pm disable-user --user 0 com.samsung.android.samsungpay.gear

(UNINSTALL AT OWN RISK)
pm uninstall -k --user 0 com.samsung.android.samsungpay.gear
pm uninstall -k --user 0 com.samsung.android.messaging
pm uninstall -k --user 0 com.samsung.android.video.wearable
pm uninstall -k --user 0 com.samsung.android.wear.voicerecorder

(KEEP)
pm uninstall -k --user 0 com.google.android.apps.wearable.retailattractloop
pm uninstall -k --user 0 com.samsung.android.watch.compass
pm uninstall -k --user 0 com.samsung.android.watch.weather
pm disable-user --user 0 com.samsung.android.bixby.agent [DONT UNINSTALL OR GPS WILL BREAK]

pm uninstall -k --user 0 com.google.android.apps.maps
pm uninstall -k --user 0 com.samsung.android.app.reminder
pm uninstall -k --user 0 com.samsung.android.watch.worldclock
pm uninstall -k --user 0 com.samsung.android.watch.cameracontroller
pm uninstall -k --user 0 com.google.android.marvin.talkback
pm uninstall -k --user 0 com.samsung.android.wearable.music
pm uninstall -k --user 0 com.google.android.wearable.assistant
pm uninstall -k --user 0 com.samsung.android.bixby.wakeup
pm uninstall -k --user 0 com.android.providers.userdictionary
pm uninstall -k --user 0 com.samsung.android.watch.worldclock

<Screen Resolution DPI Density for Google Voice>

www.reddit.com/r/WearOS/comments/10jjo0i/sideload_google_voice_on_galaxy_watch_4_wear_os_3

www.reddit.com/r/Googlevoice/comments/xxkl03/google_voice_app_for_pixel_watch/

Trying to get Google Voice to work on Galaxy Watch 4 Wear OS 3

Tried installing the apks listed here, but with respect to texting, it was totally unusable with both these versions of Google Voice (they both looked identical on the watch):
www.apkmirror.com/apk/google-inc/google-voice/google-voice-2022-10-03-481213423-release/google-voice-2022-10-03-481213423-2-android-apk-download/

www.apkmirror.com/apk/google-inc/google-voice/google-voice-2022-02-21-431956892-release/google-voice-2022-02-21-431956892-2-android-apk-download/

There are so many NO DPI variants of GV listed in the links above; I wonder if any of them would work?

Had I known there was no GV for Wear OS 3, I would have purchased a Kospet Note instead (www.youtube.com/watch?v=htJb74f5Z2o). Not having GV really is debilitating; I'm shocked it wasn't released for Wear OS 3.

As a last resort I'm considering altering the pixel density for GV, and rendering it out -- I wonder what the best density would be for GV, and if anyone has rendered a version of it
www.reddit.com/r/WearOS/comments/p3jd8t/how_to_change_the_size_of_what_you_see_on_the/

I think the ideal for Google Voice would probably be ~200. Instead of rendering GV at 200, for now, I'm just running my watch at 280 -- I LOVE IT… Tons of more screen real estate.

(ADB AppControl > Console Tab)
adb shell wm density (to get the current DPI default is "Physical density: 340")
adb shell wm density 200 (way too small for watch but a great starting point for Google Voice apk)
adb shell wm density 240 (https://youtu.be/llis8VX91tg?t=65)
adb shell wm density 250
adb shell wm density 260 (too small for watch; but a bit too big for Google Voice)
adb shell wm density 280 (----- HAPPY MIDDLE ? -----)
adb shell wm density 290 (Does something strange and is actually smaller than 280; tried 2 tests)

adb shell wm density 300 (not super small for watch; too big for Google Voice)
adb shell wm density 310
adb shell wm density 340 (Default DPI for 46mm Samsung Galaxy Watch 4 "Physical density: 340")

Great video showing different densities Galaxy Watch 4!
https://www.youtube.com/watch?v=llis8VX91tg

COMPARISON OF DENSITIES:
https://www.identity-intelligence.org/content/Watch%20Wear%20OS%203%20Galaxy%20Watch%204%2046mm%20DPI%20Density%20Test.PNG

// WATCH: SAMSUNG - WATCH FACE STUDIO

https://developer.samsung.com/watch-face-studio/download.html
https://developer.samsung.com/blog/en-us
https://forum.developer.samsung.com/
https://www.youtube.com/watch?v=8quEZE9jHu4

<URLs>

SCREEN DENSITY TESTS
https://www.reddit.com/r/WearOS/comments/10jjo0i/sideload_google_voice_on_galaxy_watch_4_wear_os_3/
https://identity-intelligence.org/content/Watch%20Wear%20OS%203%20Galaxy%20Watch%204%2046mm%20DPI%20Density%20Test.PNG

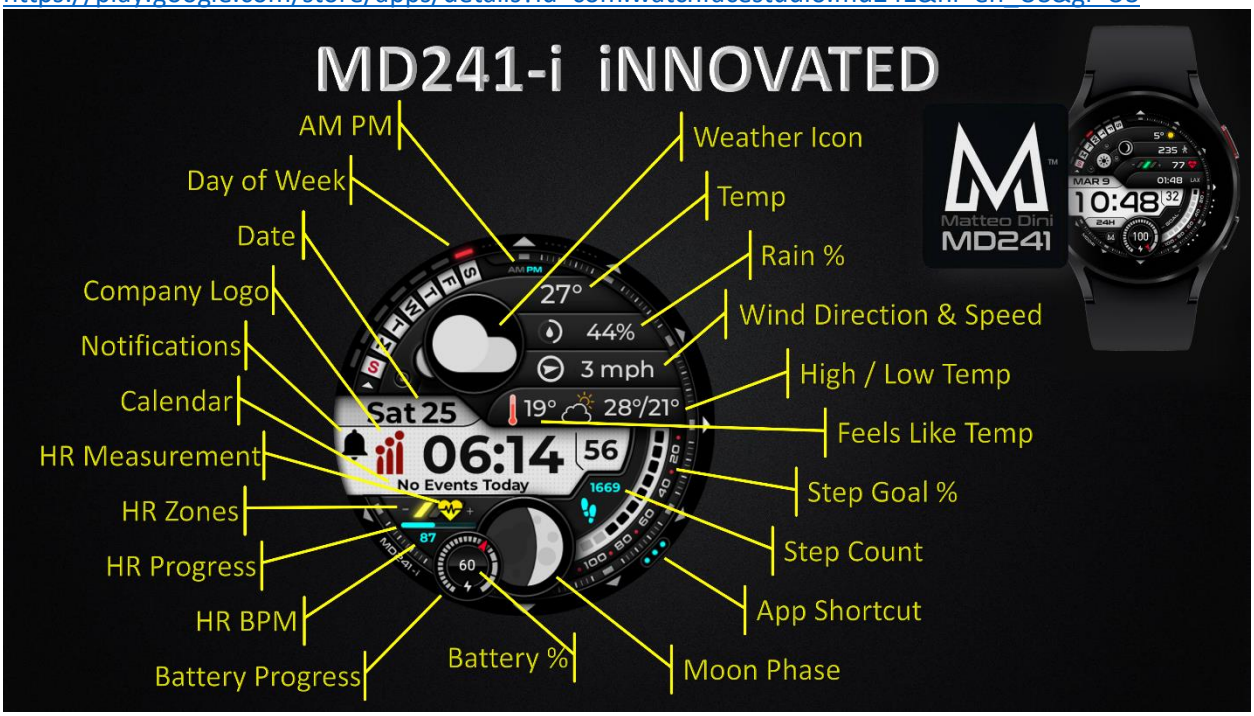
MATTEO MD294-i INNOVATED
https://www.reddit.com/r/GalaxyWatchFace/comments/111uj27/matteo_dini_md294_innovated_watch_face/
https://identity-intelligence.org/content/Watch-Face-Studio-Manual-Samsung-MD294-Sample.png
https://identity-intelligence.org/content/Watch%20Face%20MD294%20Innovated%20Side%20by%20Side.png

MATTEO MD241-i INNOVATED WEATHER
https://forum.xda-developers.com/t/ultimate-weather-matteo-dini-md241-innovated.4562093/
https://identity-intelligence.org/content/Watch-Face-Studio-Manual-Samsung-MD241-Sample.png

WATCH FACE STUDIO MANUAL
https://forum.xda-developers.com/t/mini-manual-for-samsung-watch-face-studio-text-elements-conditional-lines-etc.4558483/
https://identity-intelligence.org/content/Watch-Face-Studio-Manual-Samsung.pdf
https://identity-intelligence.org/content/Watch-Face-Studio-Manual-Samsung-TOC.png

<Matteo MD241 iNNOVATED>

This watch face was innovated with the instructions contained in this manual.
Screenshots shared with permission from Matteo:
https://play.google.com/store/apps/details?id=com.watchfacestudio.md241&hl=en_US&gl=US



<Complications>

https://developer.samsung.com/one-ui-watch-tizen/watch-face/complications.html

MAXIMUM COMPLICATIONS = 8 (Complications are blue in the left navigation pane)
https://developer.android.com/training/wearables/wfs/complications

	App Shortcut	Calendar	Assistant	Chance of Rain	Date & Weather	Feels Like Temp	Sunrise/Sunset	Weather	Moon Phase	Favorite Contact	World Clock	Calories	Distance	Floors	Steps	Battery	Notifications	Outlook Up Next	Pro Wear Next Event
Short Text	x	x			x	x	x	x	x		x	x	x	x	x	x	x		x
Monochromatic Image	x		x						x							x	x		
Ranged Value Progress Bar	x			x					x			x	x	x	x	x			
Long Text	x			x	x			x			x	x	x	x	x	x		x	x
Small Image	x								x	x									
Photo Image	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N

SHORT TEXT

- Weather / Weather Icon / Weather Feels Like - Not sure if I used same for all 3; think I used Short Text complications, but not positive
- Rain % - Ranged Value (Progress Bar)
- Outlook - Long Text
- Moon Phase - Image
- Shortcut - Image
- Battery % - Ranged Value (Progress Bar)

<Run On Device>

(Send Watch Face to Watch - top right corner of program)

PC: Follow instructions in this manual to connect to ADB AppControl [Disable BlueTooth (on phone)] > Connect Watch & PC to WiFi > Developer > Enable ADB Debugging & Debug over WiFi).

Connect with ADB AppControl and then "RUN ON DEVICE".

<Build (Render) .apks>

- Build (top right corner)
- Watch Face Studio Keystore Password use anything (I used "password")
- Default Key Store Location: C:\Users\admin\WatchFaceStudio\keystore\keystore.jks

<Unpacking Purchased Watch Images>

WATCH: Install Desired Faces

PC: ADB AppControl > Find Watch Face (e.g. search for "MD") > Check all faces you want to download > Rt-Click > Save

Saves files to:
D:\Downloads\ADB_AppControl\downloads\samsung SM-R895U\
(when installing ap it installs to Downloads so need to move to C:\Program Files\ADB_AppControl\)… then files will be here:
C:\Users\admin\AppData\Roaming\AdbAppControl\downloads\samsung SM-R895U\

Or possibly here:
C:\Program Files\ADB_AppControl\downloads\samsung SM-R895U\

.APK files unzip natively with 7ZIP (don't need to change extension to RAR or ZIP) > Extract > and extract files within (especially "base") by changing their extensions to RAR or ZIP > Might want to then search for all .png

WFP 307 Hands were 77 x 225 > Resize to 50 x 450 from the Top > Click and Drag Selection to 225 (cf. properties bar) > Hold Crosshairs there > ZOOM all the way in > M (move tool) > and DN ARROW until centered

NOTE: Seconds hand was 96 x 512; resized to 450 before cropping to 50 x 450 & for some reason worked perfectly.

<Watch Face MOCKUP CREATORS>
https://developer.samsung.com/sdp/blog/en-us/2023/01/11/new-smart-lifestyle-photos-for-watch-face-designers
https://developer.samsung.com/sdp/blog/en-us/2019/02/12/create-eye-catching-galaxy-store-listings-with-the-watch-asset-creator
https://developer.samsung.com/one-ui-watch-tizen/asset-creator.html
https://developer.samsung.com/one-ui-watch-tizen/lifestyle-photo-assets.html FOR GALAXY STORE
https://developer.samsung.com/one-ui-watch/lifestyle-photo-assets.html FOR GOOGLE PLAY STORE

<App Codes for Custom Shortcuts>

Action > Apply Action > Tap > OpenApp > Custom:

- com.samsung.android.wear.shealth (Samsung Health: Sleep, Exercise, Heart Rate, Daily Activity, Steps, Oxygen, Stress, etc.)
- com.samsung.android.shealthmonitor (Samsung Health Monitor: ECG)
- com.samsung.android.service.health (Health Platform: Unknown)
- com.google.android.wearable.healthservices (Health Services: Unknown)

<Images: Rotating Battery Meter Indicator>

https://forum.developer.samsung.com/t/is-there-any-way-to-edit-in-more-sophisticated-way-a-watch-face-designed-in-watch-face-studio/15045/4

Creating the Images:

- Make Battery Gauge & Battery Gauge Indicator Needle Exact same overall dimensions
- Set Battery Gauge Indicator Needle image to the 0% position to begin

Add Component > Shape > Image > Import both images and perfectly center them over one another.

Select Battery Gauge Indicator > Properties Pane > Rotate > Angle > Tags:

Single sweep rotation is 100 degrees):

- 360 degree sweep = **[BATT_PER]*3.6** (for MatteoMD241 gauge)
- 300 degree sweep = **[BATT_PER]*3** (for MatteoMD241 gauge)
- 180 degree sweep = **[BATT_PER]*1.8** (for half sweep)

Select Battery Gauge Indicator > Properties Pane > Rotate > Angle > Set to 0 (since Battery Gauge Indicator Needle image was set to the 0% position to begin).

Regarding Inner Pivot X / Inner Pivot Y -- since images were created identical in size (same overall dimensions):
Select Battery Gauge Indicator > A small green Pivot dot will appear > Perfectly center it over the image

<Images: Conditional Lines - Battery & Steps Goal>

[Battery]

Bring all your images in (if you want them to be in a folder: Select Them > "Group"; if you want them colored, Select Them > "Layer Color")

Menu > View > Conditional Line
Conditional Line Pane > + > Battery

On the newly created Conditional Line Battery Pane, on the horizontal line adjacent to the image you're working with, Double-Click the line where you want the image to begin (e.g. 20%)

NOTE: to easily navigate the Conditional Line horizontally: [SHIFT] + Mouse Wheel
Simply drag the start / stop time on the Battery timeline

Reference: https://developer.samsung.com/codelab/watch-face-studio/conditional-lines.html

[Steps Goal]

Menu > Project > Settings > Health > Steps

You can add the Step COUNT on watch face -- but for now not the Step PERCENT (progress bar) -- as the % will only display the Step Count that I set in “Settings” and not the S-health (as S-health isn’t linked to the watch face).

Everything else is same as Battery instructions above.

<Images: Text Elements - Heart Rate Zones>

Add Component > Text > Dimension 450 x 450 (will need to do this 4 X's for: Low, Medium, High, & Measuring)

Properties Panel (main right panel):

- Center Vertically & Horizontally (little buttons on top right) >
- Text Field [GET FROM BELOW]
- Color > Opacity (right besides the color hex) > Click & then click Tag Button that appears > Formula [GET FROM BELOW] > *Set Opacity to 0%

Text Appearance > Select Bitmap Font > "Apply Default Font Size" should be checked > Bitmap Font Settings > Default Font Size = 450 > Custom Tab > Add Text String
(Do this for each of the following and be sure to add the corresponding images with " + ")

HRLOW
HRMED
HRHIGH
HR_IS_MEASURING

[TEXT FIELDS]
HRLOW or [HRLOW]
HRMED or [HRMED]
HRHIGH or [HRHIGH]
HR_IS_MEASURING or [HR_IS_MEASURING]

[FIND YOUR PERSONAL ZONES]
https://www.heart.org/en/healthy-living/fitness/fitness-basics/target-heart-rates
Target Heart Rate = 50%-80% of Avg Max Heart Rate

https://www.myprocoach.net/calculators/hr-zones/
120 = Elliptical Machine Target Heart Rate: <=088 | 089-096 | >096
152 = Target Heart Rate (120 Starts Red Zone): <=111 | 112-122 | >122
170 = Maximum Heart Rate: <=125 | 126-137 | >137

[FORMULAS]
*NOTE: For all of these Formulas, the Opacity should be set to 0%

0 + (([HR]<=111)*100)+([HR_IS_MEASURING]*-100)

100 + ((([HR]<112)+([HR]>122))*-100)+([HR_IS_MEASURING]*-100)
0 + ((([HR]>122)*([HR]<170))*100)+([HR_IS_MEASURING]*-100)
[HR_IS_MEASURING]*100

You can remove the (HR_IS_MEASURING) if you want the text / image to stay while watchface is reading the HR data.

Show "RUN" (right panel opened with little icon on top right) > Drag Heart Rate Slider on Bottom of program to make sure all is working.

[Heart Rate Progress Bar]

Used the following:
Value: 100
Min: 100
Max: 311

THIS DIDN'T WORK:
https://forum.developer.samsung.com/t/heart-rate-meter-as-inverse-percentage-0-100-between-40-and-240/21738/2
Value: 40+240-[HR]
Max: 240
Min: 40

<Images: Text Elements - Tag Expressions & Bitmap Font>

(Custom Images for Moon Phase, AM/PM, Day of The Week, Heart Rate Zone, Etc.)

https://developer.android.com/training/wearables/wfs/tag-expression

https://developer.samsung.com/sdp/blog/en-us/2022/11/01/how-to-customize-and-enhance-text-elements-on-a-watch-face

https://developer.samsung.com/codelab/watch-face-studio/tag-expression.html#Make-use-of-tag-expressions

Make sure you're on current date / time for testing purposes!!!!

Show "RUN" (right panel opened with little icon on top right) > Drag Timeline Slider on Bottom of program to make sure all is working.

[LEDs for AM / PM]

Add Component > Digital Clock > Time > Dimension 450x450

Properties Panel (main right panel) >

- Center Vertically & Horizontally (little buttons on top right) >
- Text Field: [AMPM]
- Text Appearance > Left Aligned > Select Bitmap Font > "Apply Default Font Size" should be checked > Bitmap Font Settings > Default Font Size = 450 > Other Tab > Upload all the images

Show "RUN" (right panel opened with little icon on top right) > Drag Timeline Slider on Bottom of program to make sure all is working.

For More Advanced Ideas:
https://forum.developer.samsung.com/t/one-letter-day-of-the-week/23114/2

[LEDs for SMTWTFS]

Single = [DAY_WEEK_S]
Full = [DAY_WEEK_F]

Add Component > Digital Clock > Date

Properties Panel (main right panel):

- Dimension 450x450
- Center Vertically & Horizontally (little buttons on top right)
- Text Field: [DAY_WEEK_S]
- Text Appearance > Left Aligned > Select Bitmap Font > "Apply Default Font Size" should be checked > Bitmap Font Settings > Default Font Size = 450 > Other Tab > Upload all the images

Show "RUN" (right panel opened with little icon on top right) > Drag Timeline Slider on Bottom of program to make sure all is working.

[Wrap Long Text Complication]

https://www.reddit.com/r/WatchMaker/comments/k52evm/made_a_watchface_but_how_can_i_wrap_the_calendar/

Sample: string.sub("{c1t}", 1, 20) .. "\n" .. string.sub("{c1t}", 21, 40)

[Custom Images for Moon Phase]

Use Small Picture complication to enable Moon Phase complication from Complications Suite - Wear OS amoledwatchfaces (https://play.google.com/store/apps/details?id=com.weartools.weekdayutccomp)

See all amoledwatchfaces' complications: D:\Documents\PC PROGRAMS\0 PHONE\Watch Faces Android Wear OS Database.xlsx -- Phone & Watch Apps worksheet

https://developer.samsung.com/sdp/blog/en-us/2019/07/08/use-tag-expressions-to-create-dynamic-watch-faces

<ERROR watch face studio Build Fail Image file does not exist>

Open:
C:\Users\admin\AppData\Roaming\WatchFaceStudio\dump\main.log
Goto Very Bottom > Look for "does not exist"